AT-ADD Dataset Investigation and Preparation Log

BTP: Lightweight and Compression-Robust Audio Deepfake Detection for Edge Deployment

Purpose: Consolidated record of the dataset investigation completed before audio download and model training.

1. Why AT-ADD Track 2 Was Investigated

The project requires compact audio deepfake detectors and a strong reference model to be evaluated under leakage-safe generalization settings. The project plan assigns the baseline/generalization work to investigating how compact detectors generalize to unseen speakers and generators, with compact Mel-spectrogram CNN, compact CRNN, and one stronger learned-representation reference. ■filecite■turn0file0■L117-L123■

AT-ADD Track 2 was investigated because it contains four audio types—speech, sound, singing, and music—and explicitly supports all-type audio deepfake detection. Its metadata also provides generator identities, making it suitable for unseen-generator analysis.

2. Access and Local Setup

A Hugging Face authenticated environment was prepared inside WSL. The working location is under the BTP project directory in the WSL Linux filesystem. Access to the AT-ADD Track 2 dataset was approved, and only the metadata was downloaded at this stage.

The three metadata files downloaded were t2_train_meta.csv, t2_dev_meta.csv, and t2_eval_meta.csv. Together they are approximately 52 MB. The full audio collection is approximately 66 GB and has intentionally not yet been downloaded.

3. Metadata Schema

Each metadata CSV contains nine fields: name, type, label, source, source_id, speaker_id, generation_task, generator, and caption. The most important fields for our dataset construction are type, label, generator, speaker_id, source, and generation_task.

4. Overall Dataset Inspection

Split	Total	Speech	Sound	Singing	Music	Fake	Real
Train	146,781	49,575	39,840	36,000	21,366	113,631	33,150
Dev	91,069	49,734	19,929	16,000	5,406	71,593	19,476
Eval	229,373	144,078	28,593	24,332	32,370	184,957	44,416

Total metadata records inspected: 467,223.

5. Important speaker_id / generator Finding

Inspection of the speech metadata showed a deliberate pattern rather than ordinary random missingness: real speech has usable speaker_id values and generator='-', while fake speech has generator values and speaker_id='-'. Therefore, samples must NOT be discarded merely because one of these fields is '-'. For fake speech, generator is the relevant identity for unseen-generation analysis; for real speech, speaker_id can be used to investigate speaker overlap.

6. Speech Analysis

Split	Samples	Fake	Real	Fake generators	Usable real speakers
Train	49,575	39,576	9,999	20	3,068
Dev	49,734	39,734	10,000	20	2,105
Eval	144,078	124,078	20,000	25	6,291

Train and development contain the same 20 fake speech generators. Evaluation contains 25 different fake speech generators. Train∩Eval=0 and Dev∩Eval=0. Thus speech evaluation provides a clean unseen-generator test.

7. Speech Generator Details

The 20 train/dev speech generators include Kokoro, Glow-TTS, ProDiff, WaveNet, StarGANv2-VC, DiffSpeech, FastSpeech2, FastPitch, Parler-TTS, WaveGlow, MelGAN, Index-TTS, PortaSpeech (normal), MeloTTS, FastDiff, Tacotron2, GradTTS, Tortoise-TTS, CosyVoice, and Llasa1B. The evaluation set instead contains 25 generators including BigVGAN, HiFiGAN, DiffGANTTS, ParallelWaveGAN, OpenVoice2, IndexTTS1.5, GPTSoVITS, MBMelGAN, SeedVC1, StyleMelGAN, Llasa8B, CosyVoice3, ChatTTS, Llasa3B, StyleSpeech, F5TTS, E2TTS, CosyVoice2, StyleTTS2, SparkTTS, FireRedTTS2, Fish, IndexTTS2, StepAudioTTS, and VITS.

8. Speech Speaker-Overlap Analysis

Comparison	Overlap
Train ∩ Dev	8 real speakers
Train ∩ Eval	144 real speakers
Dev ∩ Eval	13 real speakers

The official splits are therefore not perfectly speaker-disjoint for real speech. Because fake speech has no speaker_id in this metadata, a complete binary speech experiment cannot honestly be described as perfectly speaker-disjoint. The correct strong claim is generator-disjoint evaluation, with speaker overlap separately inspected.

9. Speech Source and Generation-Task Findings

Real speech sources include LibriTTS-R, LJSpeech, Common Voice, AISHELL-3, and other sources. Evaluation real speech additionally includes EchoFake and 3D-Speaker. Fake speech includes TTS, Vocoder, Voice cloning, and Voice conversion generation tasks. The source and generation-task fields provide additional diversity for later analysis.

10. Non-Speech Generator Analysis

Type	Train fake	Train gens	Eval fake	Eval gens	Overlap
Sound	29,986	3	19,673	4	0
Singing	27,000	3	19,332	5	0
Music	17,069	4	21,874	4	0

Sound train generators: AudioLDM, AudioLDM2, Audiogen. Sound evaluation generators: Auffusion, Both-Ears-Wide-Open, MeanAudio, UniAudio2. Singing train generators: NeuCoSVC_inhouse, SeedVC, So-VITS-SVC_inhouse. Singing evaluation generators: Free-SVC, KNN-SVC, RVC-FSD, So-VITS-SVC-FSD, VEVO1.5. Music train generators: AudioLDM2, MusicGen, MusicLDM, Stable Audio open. Music evaluation generators: ACE-Step, Musictango, SSM-TTM, Uniaudio2.

11. Overall Generalization Finding

For speech, sound, singing, and music, the evaluation fake generators have zero overlap with the training fake generators. This makes generator-disjoint evaluation the strongest and most defensible generalization experiment available from the inspected metadata. It directly supports the project's question of how compact detectors behave on unseen generators.

12. Proposed Reproducible Dataset Construction

Because the original dataset is large and heavily weighted toward fake audio, a controlled subset was proposed rather than downloading everything immediately. The proposed subset is intentionally stratified by audio type, label, and generator. Fake audio is sampled equally per generator, while real audio is sampled equally per audio type where sufficient data exists.

Split	Real	Fake	Total
Train	32,000	60,000	92,000

Dev	16,000	60,000	76,000
Eval	12,536	76,000	88,536
Total	60,536	196,000	256,536

Training fake: 2,000 per generator (20 speech + 3 sound + 3 singing + 4 music = 60,000). Training real: 8,000 per audio type = 32,000. Development fake: 2,000 per generator = 60,000; development real: 4,000 per type = 16,000. Evaluation fake: 2,000 per unseen generator = 76,000. Evaluation real: speech 4,000, sound 4,000, singing 4,000, and music 536, because music real evaluation contains only 536 records in the inspected metadata.

13. Reproducibility Requirements

• Use a fixed random seed.

• Sort metadata deterministically before sampling.

• Sample independently within each required stratum.

• Save the exact selected records as train/dev/eval manifest CSV files.

• Never regenerate the dataset differently for different models.

• Store the selection script, seed, sampling rules, metadata version, and manifest checksums.

• Use the same manifests for MobileNetV3-Small, larger lightweight models, CRNN, and the stronger reference model.

• Verify the manifest counts and per-generator/type counts before downloading audio.

14. What Has NOT Been Done Yet

The full audio has not been downloaded. The deterministic manifest-generation script has not yet been written. Exact audio paths have not yet been verified for selective download. Preprocessing, Mel-spectrogram extraction, model training, robustness transformations, compression experiments, and edge benchmarking have not yet started.

15. Current Decision / Next Step

The next action should be to implement the deterministic dataset-construction script using the already downloaded metadata. The script should create train/dev/eval manifests, verify all requested strata, and produce a reproducibility summary. Only after the manifests are verified should the corresponding audio files be downloaded.

16. Team-Shareable Summary

We investigated AT-ADD Track 2 and downloaded its three metadata files without downloading the ~66 GB audio collection. The dataset contains speech, sound, singing, and music with real/fake labels. Metadata inspection found a deliberate speaker/generator structure: real speech has speaker IDs while fake speech has generator IDs. Most importantly, evaluation generators are completely unseen relative to training for speech, sound, singing, and music. Therefore, AT-ADD provides a strong generator-generalization test for our BTP. We also found that speaker overlap between official splits is small but non-zero, so we will not claim perfect speaker-disjoint binary speech evaluation. We proposed a deterministic ~256k-file subset with equal per-generator fake sampling, balanced real samples by type, fixed seed, sorted metadata, and saved manifests. The manifests will be created and verified before any large audio download.